UNIT V

Elements of Motion Control: Clutches & Brakes

internal expanding | external contracting | band | cone | energy considerations | temperature rise | friction materials

I. Introduction

A **clutch** connects / disconnects two **rotating** shafts (engine and gearbox) so that power can be transmitted or interrupted smoothly. A **brake** **slows down or stops** a rotating body by absorbing its kinetic energy as heat, or holds it at rest. Both work on **friction** between a rotating member and a stationary (brake) or another rotating (clutch) member.

Type	Clutch	Brake
Positive / mechanical	Jaw (dog) clutch, spline clutch - no slip, engage at rest or low speed	Ratchet and pawl
Friction	Disc / plate, cone, centrifugal, band, internal expanding, external contracting	Drum (internal expanding), external shoe, band, disc
Other	Hydraulic (fluid coupling), electromagnetic, one-way (sprag)	Eddy-current, regenerative, engine brake

> Automobile: **single dry plate clutch** (engine-gearbox), **drum brake** at rear (internal expanding shoes), **disc brake** at front, **band brake** in automatic transmissions, **centrifugal clutch** in scooters / mopeds, **cone clutch** in synchronisers.

> Requirements: high and **steady** μ, good heat dissipation & high temperature strength, smooth engagement, long wear life, easy adjustment.

II. Friction materials

Material (against CI / steel)	μ (dry)	Max pressure (MPa)	Max temp (°C)	Remarks
Organic / resin-bonded (non-asbestos, moulded or woven)	0.35 - 0.40	0.7 - 1.0	200 - 260	Most automotive linings; quiet, soft; fades at high temp

Material (against CI / steel)	μ (dry)	Max pressure (MPa)	Max temp (°C)	Remarks
Semi-metallic (steel wool + resin)	0.35 - 0.45	1.0 - 1.4	300 - 450	Heavy-duty, good conduction; noisy, wears disc
Sintered metal (Cu / Fe base)	0.25 - 0.40	1.5 - 3.0	400 - 600	High load, high temp, racing, heavy clutches; abrasive
Ceramic / carbon-carbon	0.30 - 0.45	> 1.0	600+	Light, stable, expensive (sports, aircraft)
Cork (in oil)	0.30 - 0.35	0.05 - 0.10	80	Wet clutches, soft engagement
Leather / cotton / wood	0.3 - 0.5	0.07 - 0.3	60 - 100	Old, belt clutches, light duty
Cast iron on cast iron (dry)	0.15 - 0.20	0.1 - 0.3	300	Brake drums, disc rotors (the rubbing partner)
Asbestos (old; now banned)	0.35 - 0.45	0.5 - 1.0	350	Banned - health hazard (asbestosis)

> Values are typical - use the data-book value for the problem. Wet (oil) μ is 0.05-0.15 lower than dry.

> **Properties wanted**: high coefficient of friction, small change with temperature (**fade resistance**), good wear resistance, thermal conductivity, strength, low compressibility, no noise/ smell, low cost, environmentally safe (no asbestos).

> **Rubbing partner**: grey cast iron (FG260), nodular iron, steel (hardened) - cast iron has good damping & heat capacity.

> **Design pressure** p_{max} is chosen from the table to give a reasonable wear life (pressure × velocity limit: $p\,v \le$ allowable).

III. Pressure distribution and torque capacity

For a friction surface the pressure distribution may be assumed **uniform pressure** (new surfaces) or **uniform wear** (after run-in: $p\,r$ = constant). Uniform-wear (the more realistic design basis) gives a lower torque capacity, so it is safer.

	Uniform pressure	Uniform wear
Axial force W (disc, radii R1 outer, R2 inner)	$W = p\,\pi\,(R1^2 - R2^2)$	$W = 2\pi\,p_{max}\,R2\,(R1 - R2)$ ($p\,r = C = p_{max}\,R2$)

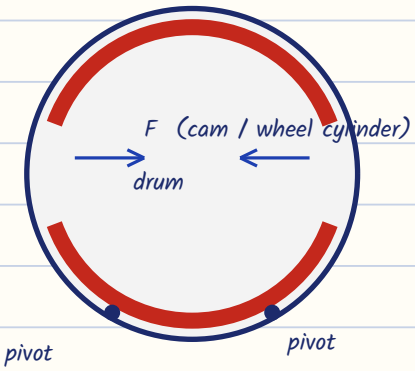
	Uniform pressure	Uniform wear
Torque per friction surface	$T = (2/3)\ \mu\ W\ (R1^3 - R2^3)/(R1^2 - R2^2)$	$T = \mu\ W\ (R1 + R2)/2 = \mu\ W\ R_m$
Mean radius R_m	$(2/3)(R1^3 - R2^3)/(R1^2 - R2^2)$	$(R1 + R2)/2$

> **Plate (disc) clutch**: torque $T = n\ \mu\ W\ R_m$ where n = number of friction surface pairs (single plate has 2 surfaces: $n = 2$).

> Typical car clutch: $R1/R2 \approx 1.2\text{-}1.3$; springs give W; $\mu \approx 0.3\text{-}0.4$; torque capacity = (1.2 - 1.5) × max engine torque (to avoid slipping).

IV. Internal expanding rim clutches and brakes

Two (or more) **shoes** lined with friction material are **expanded** outwards (by cam / wheel cylinder / centrifugal force) against the **inner surface of a rotating drum**. Used as **automobile drum brakes** and **centrifugal clutches**. The shoes are pivoted at one end (anchor), the other end is pushed by the actuating force F.



Leading shoe: friction helps to press the shoe (self-energising)

Trailing shoe: friction opposes (self-de-energising)

Torque $T = \mu\ pa\ b\ r^2\ (\cos\theta 1 - \cos\theta 2)/\sin\theta a$

b = shoe width, r = drum radius

Analysis of a pivoted shoe

> Notation: r = drum radius, b = shoe width (face), $\theta 1$, $\theta 2$ = angles of shoe start & end measured from the pivot line, a = distance of pivot from drum centre, c = distance of F from pivot (as in the figure), θa = angle at which the pressure is maximum: $\theta a = 90°$ if $\theta 2 \geq 90°$, else $\theta a = \theta 2$.

> Pressure at any point is proportional to the radial displacement of the lining, which is proportional to $\sin\theta$: **$p = pa \sin\theta / \sin\theta a$** ($pa$ = maximum pressure).

Quantity	Formula
Brake torque T (per shoe)	$T = \mu\ pa\ b\ r^2\ (\cos\theta 1 - \cos\theta 2) / \sin\theta a$
Moment of normal forces about pivot	$Mn = (pa\ b\ r\ a / \sin\theta a)\ [\ (\theta 2 - \theta 1)/2 - (\sin 2\theta 2 - \sin 2\theta 1)/4\ ]$
Moment of friction forces about pivot	$Mf = (\mu\ pa\ b\ r / \sin\theta a)\ [\ r\ (\cos\theta 1 - \cos\theta 2) - a\ (\sin^2\theta 2 - \sin^2\theta 1)/2\ ]$
Actuating force F	Leading (self-energising) shoe: $F = (Mn - Mf)/c$; trailing (de-energising) shoe: $F = (Mn + Mf)/c$
Self-locking	when $Mn \leq Mf$ ($F \leq 0$): brake grabs on its own. Avoid by designing $Mn > Mf$ (smaller a, μ)
Pivot reactions	$Rx = \int$ components of p and friction; resultant $= \sqrt{(Rx^2 + Ry^2)}$ for design of the anchor pin
Zero-friction-moment pivot (symmetrical shoe)	$a = 4\ r \sin\alpha / (2\alpha + \sin 2\alpha)$ (α = half angle of shoe) -> $Mf = 0$, wear even, no self-energising

> **Leading shoe**: friction drags the shoe into the drum (self-energising, needs less F, wears more, higher pressure). **Trailing shoe**: friction pulls it away. For the same F the leading shoe gives more torque - a duplex / 'two leading shoe' brake uses both leading in forward motion.

> Total brake torque = T(leading) + T(trailing), with the pressure pa of the trailing shoe less than that of the leading shoe for the same F.

> Heat dissipation: drum brake dissipates heat through the drum only; fade at high temperature is the main limit.

Example 1 – internal expanding brake (two shoes)

> Drum radius r = 150 mm, shoe width b = 40 mm, θ1 = 10°, θ2 = 120° (so θa = 90°), pivot distance a = 150 mm, μ = 0.3, F acts at c = 250 mm, leading shoe pa = 0.8 MPa.

> Torque (leading) $T = \mu\, pa\, b\, r^2(\cos 10° - \cos 120°) = 0.3 \times 0.8 \times 40 \times 150^2 \times 1.4848 =$ **320,700 N·mm**.

> $Mn = pa\, b\, r\, a\, [\,(\theta 2 - \theta 1)/2 - (\sin 240° - \sin 20°)/4\,] = 0.8 \times 40 \times 150 \times 150 \times 1.262 = 908{,}600$ N·mm ; $Mf = 0.3 \times 0.8 \times 40 \times 150\, [150 \times 1.4848 - 150 \times (0.75 - 0.0302)/2] = 243{,}000$ N·mm.

> $F = (Mn - Mf)/c = (908{,}600 - 243{,}000)/250 =$ **2662 N**. Trailing shoe with the same F: $pa' = F\, c/((Mn + Mf)/pa) = 0.462$ MPa -> $T' = 185{,}400$ N·mm. Total brake torque = 320,700 + 185,400 = **506,100 N·mm ≈ 506 N·m**.

Centrifugal clutch (internal expanding clutch)

> Shoes carried on a spider fixed to the driving shaft; they fly outward against a drum of the driven member when the speed exceeds the **engagement speed** (about 0.7 of running speed), held back by garter springs.

> Centrifugal force per shoe $Fc = m\, \omega^2\, r_g$; spring force $Fs = m\, \omega 1^2\, r_g'$ at engagement; net normal force on drum $R_N = Fc - Fs$.

> Torque transmitted $T = n\, \mu\, R_N\, r$ (n = number of shoes). Used in scooters, mopeds, starters, chain saws - automatic, no pedal.

> Design of drum / shoes: from the power: $T = 9.55 \times 10^6\, P/N$; contact pressure $p = R_N / (b\, l) \leq$ allowable (0.1 MPa for cork/ leather, 0.3 MPa for asbestos-free).

V. External contracting rim clutches and brakes

Shoes press against the **outside** of the rotating drum (contracting). Types: **single block (short shoe)**, **pivoted block (long shoe)**, **double block** (two blocks acting on the drum from opposite sides - balances the shaft load) and the pivoted long shoe is most used in industry / railways.

1. Single block (short shoe) brake

> Shoe subtends a small angle (< 45°) so the pressure is almost uniform; normal force R_N on the drum, friction force $Ft = \mu\, R_N$; torque $T_B = \mu\, R_N\, r$.

> Lever: force F at distance l from the fulcrum, shoe at distance x along the lever, line of the friction force at a perpendicular distance a from the fulcrum:

Case	Actuating force F
Friction force through fulcrum (a = 0)	F = R_N x / l - same for both directions
Drum rotation makes friction force assist (self-energising)	F = R_N (x - μ a) / l
Drum rotation opposes (friction resists the applied force)	F = R_N (x + μ a) / l
Self-locking condition	x ≤ μ a (F becomes zero or negative) -> a < x/μ must be kept

> **Block pressure**: p_b = R_N/(b l_b) ≤ allowable (0.4-1 MPa); drum wrap angle 2θ; for angle ≥ 60° the equivalent coefficient μ' = 4μ sin(θ/2)/(θ + sin θ) is used (wedge effect).

> **Heat generated** per second Hg = μ R_N v (W); block area A_b = b l_b; check p v.

> Single block loads the shaft with side thrust R_N -> use a **double block** (equal & opposite).

2. Pivoted (long) shoe external brake

> Pressure p = pa sin θ / sin θa again; same torque equation T = μ pa b r² (cos θ1 - cos θ2)/sin θa for both clutch and brake; Mn and Mf have the same form (sign of Mf reverses with direction of rotation).

> Choose the pivot position a = 4 r sin α/(2α + sin 2α) so that Mf = 0 -> uniform wear, no self-locking, same torque in both directions.

> Pivot reaction (anchor load) = vector sum of normal and friction components - needed to size the pivot pin.

Brake / clutch type	Torque equation	Notes
Internal expanding shoe	T = μ pa b r² (cos θ1 - cos θ2)/sin θa	Drum brake; centrifugal clutch
External long shoe (pivoted)	same	Railway brake, press clutch
Single block (short)	T = μ R_N r	Hand brake, small hoist
Band	T = (T1 - T2) r; T1/T2 = e^(μθ)	Winches, chain saws, auto transmission
Cone	T = μ W R_m / sin α	Synchroniser, heavy clutch
Plate / disc	T = n μ W R_m	Car clutch, disc brake: T = 2 μ F R_m

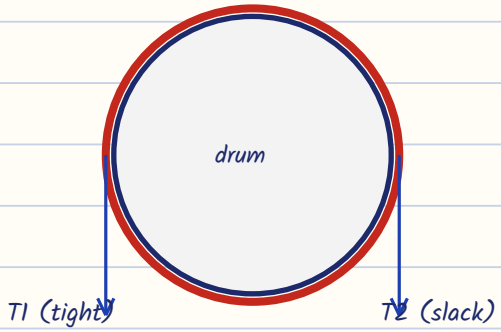
Example 2 – single block brake

> Drum 300 mm diameter, brake torque required T = 150 N·m = 150,000 N·mm, μ = 0.25, x = 250 mm, l = 600 mm, a = 40 mm (friction force offset), self-energising direction.

> $R_N = T/(\mu r) = 150{,}000/(0.25 \times 150) =$ **4000 N** ; $F = R_N (x - \mu a)/l = 4000\ (250 - 10)/600$ = **1600 N**. (If the rotation is reversed: $F = 4000 \times 260/600 = 1733$ N.)

> Self-locking if $x \le \mu a$: here 250 > 10, so not self-locking ✓.

VI. Band type clutches and brakes



$T1 / T2 = e^{(\mu \theta)}$

Brake torque $T_B = (T1 - T2)\ r$

Max band stress: $\sigma = T1 / (b\ t)$

Max pressure $p = 2\ T1 / (b\ D)$

Band lined with friction material; lever gives mechanical advantage.

> A flexible steel band (lined with friction material) is wrapped around the drum through angle θ; tightening the band produces a tension difference.

> **Euler's belt friction**: $T1 / T2 = e^{(\mu \theta)}$; brake torque $T_B = (T1 - T2)\ r$ (r = effective radius = drum radius + band thickness/2).

> Band stress $\sigma = T1/(b\ t) \le$ 80–100 MPa; maximum pressure at tight end $p_{max} = 2\ T1/(b\ D) \le 0.6$ MPa (lining).

> Wrap angles up to 270°–300°; $\mu \approx 0.25$–0.4 (lining on CI drum).

Band brake type	Lever relation (F at distance l from fulcrum)	Remarks
Simple	$F = T2\ a / l$ (slack end at a from fulcrum, tight end at fulcrum)	Self-energising in one direction, not in the other
Differential	$F\ l = T2\ b - T1\ a$ (ends on either side of fulcrum)	Self-locking if $b/a \le e^{(\mu \theta)}$; high mechanical advantage; risk of grab

Band brake type	Lever relation (F at distance l from fulcrum)	Remarks
Band and block	Wooden / lined blocks on the band; T1/T2 = ((1 + μ tan(φ/2))/(1 - μ tan(φ/2)))^n	Heavy cranes; blocks replaceable

Example 3 – simple band brake

> Drum 300 mm dia, wrap 270° (4.71 rad), μ = 0.3, required brake torque 400 N·m. e^(μθ) = e^(1.414) = 4.11.

> T1 - T2 = T/r = 400,000/150 = 2667 N ; T1 = 2667 × 4.11/3.11 = **3525 N** ; T2 = 858 N.

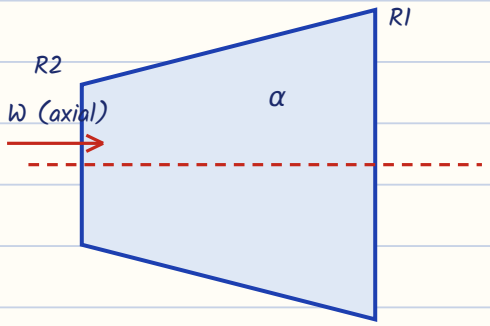
> Band width from p_max = 2 T1/(b D) = 0.6 N/mm² -> b = 2 × 3525/(0.6 × 300) = 39.2 -> **b = 40 mm**. Lever: a = 50 mm, l = 500 mm -> F = T2 a/l = 858 × 50/500 = **85.8 N**.

Band clutch

> Same theory: friction band tightened on a rotating drum connects the two shafts; torque = (T1 - T2) r. Actuated by a lever & toggle; self-energising in one direction.

> Used in earth-moving machines, winches, and **bands of automatic transmissions** (band holds the planetary drum).

VII. Cone clutches and brakes



α = semi-cone angle (10°–15°)

Rm = (R1 + R2)/2

T = μ W Rm / sin α

Engaging force F = W (sin α + μ cos α)

Cone gives a wedging action:

less axial force than a plate clutch.

Two conical surfaces (male cone, female cone) engage under axial force W. The **wedge action** multiplies the friction force (normal force N = W/(sin α + μ cos α)), so a cone clutch needs less axial force than a plate clutch of the same torque - but is bulky and may jam. Semi-cone angle α is 10°–15° (must be larger than the friction angle $\tan^{-1}\mu$ to release easily).

Case	Axial force W	Torque T
Uniform pressure p	$W = p\,\pi\,(R1^2 - R2^2)$	$T = (2/3)\,\mu\,W\,(R1^3 - R2^3) / [\,\sin\alpha\,(R1^2 - R2^2)\,]$
Uniform wear (p_{max} at R2)	$W = 2\pi\,p_{max}\,R2\,(R1 - R2)$	$T = \mu\,W\,R_m / \sin\alpha$, $R_m = (R1 + R2)/2$
Face width b & dimensions	$b = (R1 - R2)/\sin\alpha$	Normal pressure $p = W/(2\pi\,R_m\,b\sin\alpha)$
Engaging force	$F = W(\sin\alpha + \mu\cos\alpha)$ (force needed to engage while rotating)	Disengaging force $Fd = W(\mu\cos\alpha - \sin\alpha)$ (if positive)

Example 4 – cone clutch

> $R1 = 120$ mm, $R2 = 100$ mm, $\alpha = 12.5°$, $\mu = 0.25$, $T = 300$ N·m (uniform wear). $R_m = 110$ mm.

> $W = T\sin\alpha/(\mu\,R_m) = 300{,}000 \times 0.2164/(0.25 \times 110) =$ **2361 N**; face width $b = (R1 - R2)/\sin\alpha = 20/0.2164 = 92$ mm.

> Pressure $p = W/(2\pi\,R_m\,b\sin\alpha) = 2361/(2\pi \times 110 \times 92 \times 0.2164) = 0.17$ N/mm² ✓; engaging force $F = W(\sin\alpha + \mu\cos\alpha) = 2361\,(0.2164 + 0.244) =$ **1087 N**.

Single plate clutch and disc brake (automobile) – quick relations

> **Single plate clutch**: two friction faces ($n = 2$) lined with organic material: $T = 2\,\mu\,W\,R_m$ (uniform wear). Spring force W = (6-9 springs). Capacity = 1.3 × engine torque.

> **Disc brake**: caliper squeezes two pads on a rotor: $T = 2\,\mu\,F\,R_m$ (F = clamp force each side, R_m = mean pad radius); better heat dissipation & no self-energising (stable, less fade) vs drum brake.

> Drum vs disc: drum is self-energising (high gain, cheap, parking brake) but fades & heavy; disc has uniform torque, ventilated rotor, quick recovery.

VIII. Energy considerations

All the energy of a brake (and the slip energy of a clutch) is converted into **heat**. The brake or clutch must be designed so that this heat does not overheat the friction surfaces.

Case	Energy absorbed E (J)
Rotating body slowed from ω1 to ω2	$E = ½\,I\,(\omega1^2 - \omega2^2)$, I = moment of inertia (kg m²)
Translating body (vehicle) from v1 to v2	$E = ½\,m\,(v1^2 - v2^2)$

Case	Energy absorbed E (J)
Lowered load (hoist) through height h at constant speed	$E = m\ g\ h$ (potential energy) + kinetic part
Combined	$E = ½\ I\ (\omega 1^2 - \omega 2^2) + ½\ m\ (v1^2 - v2^2) + m\ g\ h$ (+ work done by friction etc.)
Brake torque, time	$T = I\ (\omega 1 - \omega 2) / t$ (constant deceleration); stopping angle $\theta = \omega 1\ t/2$; $E = T\ \theta$
Clutch engagement (driven inertia I2 from rest to ω1 with constant torque T)	Slip time $t = I2\ \omega 1 / T$; energy dissipated = $½\ I2\ \omega 1^2$ (equals the KE gained by the driven part); mean power = E/t

> **Equivalent inertia** of a vehicle at the wheel: $I_{eq} = m\ r_w^2$ + rotating parts; for brakes on 'n' wheels the energy is shared according to the weight transfer (front 60-70%).

> Frequent clutch operation (starting) should have short slip time but also gentle engagement: engagement time 0.5-1 s for cars; greater energy for heavy trucks & large inertia.

IX. Temperature rise

$$\Delta t = E / (m\ c) \quad \text{(single stop)} \quad ; \quad \text{steady: } Hg = \mu\ p\ v\ A = Hd = h\ A\ (t - t0)$$

> m = mass of the parts that absorb heat (drum/ rotor, kg); c = specific heat (cast iron / steel ≈ 460-500 J/kg°C).

> **Single application**: all heat absorbed by the drum (short time, no loss): $\Delta t = E/(m\ c)$. Limit $\Delta t \approx 100$ °C for single stop; surface temperature ≤ 250-350 °C (drum), ≤ 600 °C (disc).

> **Continuous (cyclic) braking**: heat generated per second $Hg = Ft\ v = \mu\ p\ A\ v$; heat dissipated $Hd = h\ A\ (t - t0)$ with h = 15-30 W/m²°C (still air) to 60-100 W/m²°C (ventilated, moving vehicle) -> equilibrium temperature rise $t - t0 = \mu\ p\ v / h$.

> **p v rule**: the product p v (MPa × m/s) limits wear and temperature: for organic lining p v ≤ about 1-2 MPa m/s (dry); in oil 2-3.

> Cooling: ribs/fins on the drum, ventilated (drilled / vane) disc rotors, air ducts, oil spray (wet clutch), large drums with high thermal mass; avoid brake fade (loss of μ at high temp) by choosing materials.

Example 5 – energy and temperature rise in a vehicle brake

> Car of mass 1200 kg stopped from 80 km/h (22.2 m/s). $E = ½ m v^2 = 0.5 \times 1200 \times 22.2^2 =$ **296 kJ**. Four brakes (equal): 74 kJ each (front brakes actually take 60–70%).

> Each disc/ drum has m = 6 kg of cast iron, c = 500 J/kg°C: $\Delta t = 74{,}000/(6 \times 500) =$ **24.7 °C** per stop (rises sharply with repeated stops: 5 stops in succession -> about 120 °C if no cooling).

> Time to stop with deceleration 0.6 g = 5.9 m/s^2: t = 22.2/5.9 = 3.8 s ; mean braking power = 296 kJ/3.8 s = 78 kW (all 4 brakes) - therefore good cooling is vital.

Example 6 – clutch slip energy

> Driven inertia I2 = 0.8 kg m^2 picked up from rest to engine speed 2000 rpm (209 rad/s) by clutch torque 120 N·m.

> Slip time $t = I2\,\omega I/T = 0.8 \times 209/120 = 1.4$ s ; energy dissipated $E = ½\ I2\ \omega I^2 = 0.5 \times 0.8 \times 209^2 =$ **17.5 kJ** ; mean heating rate = 17.5/1.4 = 12.5 kW.

> If the pressure plate + flywheel face mass is 8 kg (c = 500): $\Delta t = 17{,}500/(8 \times 500) =$ **4.4 °C** - fine for one start; repeated starts need checking.

X. Quick revision & important questions

Formulas to remember

> Pressure: uniform pressure $T = (2/3)\ \mu\ W\ (R1^3 - R2^3)/(R1^2 - R2^2)$; uniform wear $T = \mu\ W\ R_m$ ($R_m = (R1 + R2)/2$).

> Shoe: $T = \mu\ pa\ b\ r^2\ (\cos \theta1 - \cos \theta2)/\sin \theta a$; $F = (Mn \mp Mf)/c$; self-locking if $Mn \le Mf$.

> Block: $F = R_N\ (x \mp \mu\ a)/l$; $T = \mu\ R_N\ r$. Band: $T1/T2 = e^{\wedge}(\mu\theta)$; $T = (T1 - T2)\ r$; $F = T2\ a/l$ (simple).

> Cone: $T = \mu\ W\ R_m\ /\ \sin \alpha$; $F = W\ (\sin \alpha + \mu \cos \alpha)$. Energy: $E = ½\ I\ (\omega1^2 - \omega2^2) = ½\ m\ v^2$; $\Delta t = E/(m\ c)$.

1. Compare clutches and brakes; list the friction materials with μ and p_max. (5 M)
2. Derive the torque capacity of a clutch for uniform pressure and uniform wear. (10 M)

3. Analyse an internal expanding shoe brake: torque, actuating force and self-locking. (10 M)
4. Explain the external contracting shoe brake: single block and pivoted long shoe. (10 M)
5. Derive the relation for a band brake; explain simple and differential band brakes. (10 M)
6. Design a cone clutch to transmit given power (find dimensions, axial force). (10 M)
7. Explain energy considerations and temperature rise in clutches/ brakes. (5 M)
8. A vehicle brake absorbs given energy - find the temperature rise. (5 M)